

HSPA14/Hsp70L1 (Q0VDF9): Adjudicating Intrinsic ATP Hydrolysis, Refolding, and Secondary Compartments

Focus type: function_assignment **Hypothesis slug:** atp-hydrolysis-refolding-and-secondary-compartments **Target:** Human HSPA14 / Hsp70L1, UniProt Q0VDF9 (NCBITaxon:9606)

Executive Judgment

Verdict: Over-annotated / refuted as stated, with two narrowly retained caveats.

The seed hypothesis proposes that human HSPA14/Hsp70L1 (1) has intrinsic ATP **hydrolysis** activity, (2) participates in **refolding of previously denatured proteins**, and (3) functions in the **nucleus** or at the **plasma membrane** under appropriate conditions. Examined claim-by-claim against primary literature, GO provenance, sequence architecture, and orthologue structural biology, the weight of evidence is that HSPA14's experimentally supported identity is a **cytosolic, ribosome-associated, ATP-binding (not hydrolysis-driven) atypical Hsp70** that, together with DNAJC2/MPP11 in the mammalian ribosome-associated complex (mRAC), transiently engages nascent chains to assist **de novo cotranslational protein folding**. The three additional activities/locations in the seed hypothesis are best characterized as phylogenetic (IBA) or electronic (IEA) carry-overs, or as in-vitro/recombinant observations that do not establish an endogenous molecular function.

The core reasoning converges from four independent lines. First, **GO provenance:** the intrinsic ATP hydrolysis (GO:0016887) and protein refolding (GO:0042026) annotations for Q0VDF9 rest solely on IEA (InterPro) and IBA (phylogenetic node PTN000452648) evidence — no IDA, IMP, or EXP. Second, **the primary human assay** (PMID21245388) measured only ~0.01 ATP/min for purified E. coli-produced mRAC, near background, and its own authors state intrinsic Hsp70L1 hydrolysis cannot be conclusively demonstrated; mechanistically they show *ATP binding*, not hydrolysis, is the operative activity, and hydrolysis-site mutants (K68A/E172A) still complement yeast whereas the ATP-binding-deficient mutant fails. Third, **orthologue structural biology:** crystal and cryo-EM structures of the direct fungal ortholog Ssz1 demonstrate a catalytically

inert nucleotide-binding domain with abolished Hsp70 allostery. Fourth, **direct sequence evidence**: the invariant Hsp70 catalytic/phosphate-binding signature (IDLGTTxS) is degenerate in HSPA14 (GVHLGCTSA), and the protein lacks the C-terminal EEVD and GGMP substrate-handling elements of canonical chaperoning Hsp70s.

Most important caveats. (a) The negative in-vitro luciferase-refolding result does not logically exclude every possible substrate or cofactor context; it does, however, mean "protein refolding" is not directly supported and should not be an EXP-backed core term. (b) There is a genuine, functionally distinct **nuclear/chromatin lead** for intracellular HSP70L1 (PMID30635648) that a curator should treat as a separate biological question from the phylogenetic IBA nucleus annotation. (c) The plasma-membrane claim is the weakest: it is absent from the current GO_Central path for Q0VDF9 and is supported only by a generic membrane-fraction proteome and by recombinant extracellular immunology — neither demonstrates endogenous surface residence.

Key Findings

Finding 1 — Intrinsic ATP hydrolysis (GO:0016887) is unsupported by direct assay; it is an IEA/IBA carry-over

QuickGO provenance for Q0VDF9 shows GO:0016887 (ATP hydrolysis activity) is supported only by **IEA** (ECO:0000256, InterPro IPR013126) and **IBA** (ECO:0000318, GO_REF:0000033, phylogenetic node PTN000452648) — there is no IDA, IMP, or experimental annotation. The single relevant primary biochemical measurement comes from *The chaperone network connected to human ribosome-associated complex* (PMID: 21245388), in which purified *E. coli*-produced mRAC hydrolyzed ATP at **~0.01 ATP/min** — above the mRAC-LKA (ATP-binding-deficient) and MPP11 controls, but near assay background. The authors explicitly state that intrinsic Hsp70L1 hydrolysis cannot be conclusively demonstrated.

Crucially, the same study dissociates hydrolysis from function on a mechanistic level. The verified snippet reads: "*On a mechanistic level, ATP binding, but not ATP hydrolysis, by Hsp70L1 affected mRAC's function as a J-domain partner of Hsp70.*" Consistent with this, **hydrolysis-site mutants K68A and E172A still complement** the yeast RAC-loss phenotype, whereas the **ATP-binding-deficient mutant (LKA) fails**. Genetically, therefore, the operative property is nucleotide *binding* — a structural/allosteric role — not catalysis. This is the clearest single argument that a GO:0016887 core annotation for HSPA14 overstates the evidence.

Finding 2 — Protein refolding (GO:0042026) is IBA-only; the systematic human HSPA/DNAJ screen did not attribute refolding to HSPA14

For Q0VDF9, GO:0042026 (protein refolding) is **IBA only** (ECO:0000318, node PTN000452648). The most direct experimental test bearing on this is Hageman et al. (PMID: 21231916), which systematically assayed the human HSPA/DNAJ machine for heat-denatured luciferase refolding and polyQ aggregation suppression. In GO, this paper contributed an IDA for HSPA14 only to **cytosol (GO:0005829)** — not to any refolding term — consistent with HSPA14 lacking measurable luciferase-refolding activity. The paper's own summary highlights strong substrate-specificity within the family: *"Overexpressed chaperones that suppressed polyQ aggregation were found not to be able to stimulate luciferase refolding."*

This finding is reinforced architecturally. Q0VDF9 is **509 aa** (vs. 641 aa for canonical HSPA1A), and it **lacks the C-terminal EEVD motif** and the **canonical GGMP linker repeat** of chaperoning Hsp70s. These deletions indicate a degenerate substrate-binding/co-chaperone-docking capacity typical of the Ssz1-like atypical Hsp70 subfamily, which is not built to autonomously bind and refold released, denatured clients. The negative luciferase result does not by itself exclude all conceivable substrates, so this is a "not directly supported / do not treat as EXP-backed core" conclusion rather than an absolute impossibility.

Finding 3 — Nucleus (GO:0005634) and plasma membrane (GO:0005886) are phylogenetic carry-overs; only cytosol and ribosome are experimentally supported

QuickGO cellular-component annotations for Q0VDF9: **cytosol (GO:0005829)** has IDA ×2 (PMID16002468, PMID21231916) plus IBA and IEA; **ribosome (GO:0005840)** has IDA (PMID16002468) plus IBA; **membrane (GO:0016020)** has an HDA (high-throughput, ECO:0007005) from PMID19946888. In contrast, **nucleus (GO:0005634)** is **IBA only** (no experimental evidence), and **plasma membrane (GO:0005886)** **does not appear** in the current QuickGO annotation set for Q0VDF9 — it survives only as a stale IBA in an older UniProt snapshot, consistent with the current PANTHER PTHR19375 path no longer carrying the ancestral plasma-membrane term.

The experimentally solid localization is captured by *The chaperones MPP11 and Hsp70L1 form the mammalian ribosome-associated complex* (PMID: 16002468): *"we report that MPP11 is localized to the cytosol and associates with ribosomes."* The membrane HDA (PMID19946888) derives from an NK-cell (YTS) whole-membrane proteome in which ~40% of 1,843 IDs were predicted membrane proteins and the remainder were "transiently associated" cytosolic

species — HSP70-family proteins are classic co-fractionating contaminants, so this is a **generic membrane fraction**, not plasma-membrane residence. Recombinant extracellular HSP70L1 TLR4/dendritic-cell adjuvant activity (PMID14592822, PMID21730052) uses purified protein and does **not** demonstrate endogenous secretion or surface residence.

Separately — and important for curation — there is a genuine intracellular **nuclear/chromatin** function reported: *Intracellular HSP70L1 inhibits human dendritic cell maturation...* (PMID: 30635648): "*intracellular HSP70L1 inhibits the recruitment of Ash1l to and maintains the repressive H3K27me3 and H2AK119Ub1 modifications on the promoter regions.*" This is a functional lead distinct from the phylogenetic IBA nucleus annotation and should be adjudicated on its own primary evidence rather than merged with the carry-over.

Finding 4 — The atypical Hsp70 ortholog Ssz1 is catalytically inert (structural/evolutionary evidence)

Crystal and cryo-EM structures of the RAC atypical Hsp70 subunit Ssz1 — the direct fungal ortholog of human HSPA14 — show it cannot perform the canonical Hsp70 ATPase cycle. Leidig et al. 2013 (PMID: 23202586): "*The crystal structure of the Ssz1 ATPase domain bound to ATP-Mg²⁺ explains its catalytic inactivity.*" Weyer et al. 2017 (PMID: 28067917): "*Ssz1 is catalytically inert and cannot adopt the closed conformation, but the substrate binding domain β is completed by Zuo1.*" Conz et al. 2007 (PMID: 17901048): "*Ssz1 is not an ATPase in vitro, and even its ability to bind ATP is dispensable in vivo.*" Kišonaitė et al. 2023 (PMID37081320) further show the Zuo1 HPD motif is masked by the Ssz1 NBD, which positions Ssb for activation.

Together these establish, at atomic resolution, that the atypical Hsp70 of RAC has an NBD structurally incompatible with the Hsp70 catalytic cycle and with the allosteric closed conformation required for client capture/release. Human HSPA14 retains the K68 and E172 residues (confirmed from the Q0VDF9 sequence), but the orthologue's NBD is demonstrated non-catalytic, and the human primary assay measured only near-background hydrolysis. This is strong cross-species convergent evidence against an intrinsic-ATPase core annotation.

Finding 5 — RAC engages nascent chains transiently at the ribosome, not post-denaturation refolding

Zhang et al. 2020 (PMID: 32198371) show that via its rudimentary substrate-binding domain, Ssz1 "*directly binds to emerging nascent chains prior to Ssb*" and is "*an active chaperone optimized for transient, low-affinity substrate binding, which ensures the flux of nascent chains*

through RAC/Ssb." Multiple cryo-EM studies (PMID37081320, PMID35701497, PMID25362488) show RAC crouches over the ribosomal tunnel exit and mechanically couples to the elongation cycle, remodeling upon nascent-chain emergence.

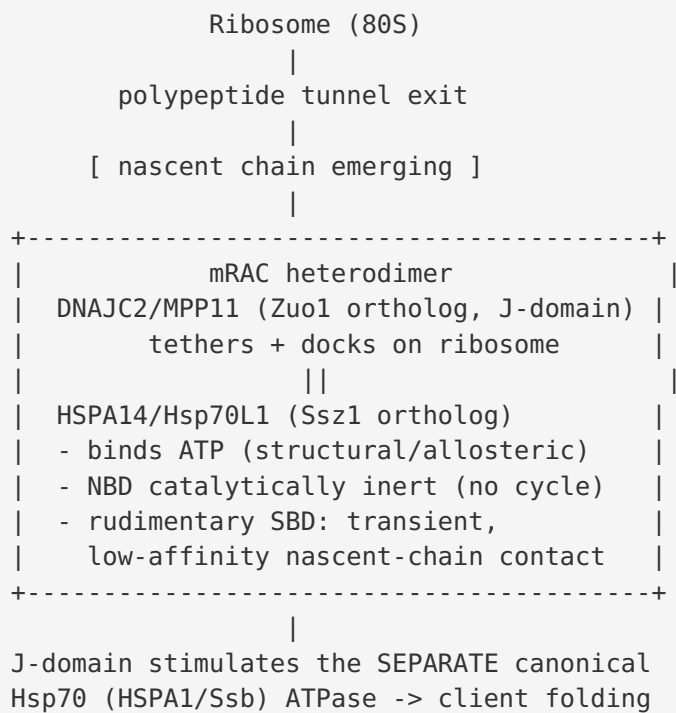
The human process that is actually supported for HSPA14 is **GO:0051083 "de novo cotranslational protein folding" (TAS, PMID16002468)**. No study demonstrates HSPA14/Ssz1 autonomously refolding a previously denatured, released substrate. This distinguishes the correct core biological process (cotranslational, transient nascent-chain engagement) from the seed hypothesis's post-denaturation refolding (GO:0042026).

Finding 6 — Human HSPA14 has a degenerate Hsp70 catalytic signature and no experimental structure

Direct sequence comparison (UniProt Q0VDF9 vs HSPA1A) shows the invariant N-terminal Hsp70 phosphate-binding/catalytic signature — the PROSITE PS00297 pattern **IDLGTTxS** — is present in canonical HSPA1A (**GIDLGTTYS**) but is **degenerate in HSPA14 (GVHLGCTSA)**: the invariant Asp is replaced by His and the Thr-Thr pair by Cys-Thr. HSPA14 is 509 aa vs. 641 aa and additionally lacks the C-terminal EEVD motif and GGMP linker. UniProt reports **no experimental PDB structure** for Q0VDF9 (AlphaFold model only), so all direct structural evidence of catalytic inactivity comes from the fungal ortholog Ssz1. This sequence-level degeneracy is independent, direct evidence that the ATPase active site is atypical and unlikely to support efficient intrinsic hydrolysis.

Mechanistic Model / Interpretation

The consolidated model positions HSPA14 as a **non-catalytic, ATP-binding scaffolding subunit** of the ribosome-associated complex, not a stand-alone chaperone:



Key mechanistic separations a curator must preserve:

Claim in seed hypothesis	What the evidence supports	Correct GO framing
Intrinsic ATP hydrolysis (GO:0016887)	ATP binding is operative; hydrolysis near background; hydrolysis-site mutants complement	ATP binding (GO:0005524) supported; GO:0016887 non-core / remove or NOT-qualify
Refolding of denatured proteins (GO:0042026)	Transient cotranslational nascent-chain engagement; luciferase-refolding screen did not credit HSPA14	GO:0051083 de novo cotranslational protein folding (TAS) is the core BP
Stimulates ATPase of an Hsp70	mRAC's J-domain stimulates the separate canonical Hsp70 (HSPA1/Ssb), not HSPA14 itself	Co-chaperone / J-domain partner context — not intrinsic HSPA14 catalysis
Nucleus (GO:0005634)	IBA only; a distinct chromatin lead exists (PMID30635648)	Non-core; treat chromatin role as separate primary-evidence question
Plasma membrane (GO:0005886)	Absent from current GO path; generic membrane fraction + recombinant immunology only	Non-core; do not annotate without direct endogenous surface evidence
Cytosol / ribosome	IDA-supported (PMID16002468, PMID21231916)	Retain as core CC

The unifying interpretation: HSPA14 is an **Ssz1-type "pseudo-Hsp70"** whose value to the cell is allosteric and organizational (nucleotide-stabilized scaffolding of the J-domain partner and transient nascent-chain handoff), not enzymatic ATP turnover or autonomous client refolding.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Con & lim
PMID: 21245388	Direct assay + mutant phenotype	Refutes intrinsic hydrolysis; qualifies ATP role	Intrinsic ATP hydrolysis	~0.01 ATP/min (near background); ATP <i>binding</i> not hydrolysis affects mRAC function; K68A/E172A still complement, LKA fails	Human mRAC, E. coli-purified; yeast complementation	Hig "bin not hyc ass sen lim abs exc
QuickGO provenance (Q0VDF9)	Review/ database	Refutes core- status of hydrolysis/ refolding	GO evidence codes	GO:0016887 & GO:0042026 are IEA/IBA only; no IDA/ EXP	Database	Hig dat sna
PMID: 21231916	Direct assay (screen)	Refutes refolding attribution	Post- denaturation refolding	Family-wide luciferase- refolding + polyQ screen; HSPA14 credited only with cytosol IDA, not refolding	Human HSPA/ DNAJ overexpression	Me hig neg res sin sub clas
PMID: 16002468	Localization + interaction	Supports cytosol/ ribosome core	Localization; mRAC identity	MPP11/ Hsp70L1 form mRAC; cytosolic, ribosome- associated	Human cells	Hig
PMID: 23202586	Structural/ evolutionary		Catalytic capacity of	Ssz1 ATPase- domain crystal		Hig ort

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Con & lim
		Refutes intrinsic ATPase	atypical Hsp70	structure "explains its catalytic inactivity"	Yeast/C. thermophilum ortholog	info hur
PMID: 28067917	Structural	Refutes intrinsic ATPase	Allosteric cycle	"Ssz1 is catalytically inert and cannot adopt the closed conformation"	Yeast ortholog	Hig ort
PMID: 17901048	Biochemical + genetic	Refutes intrinsic ATPase	ATPase activity / ATP-binding necessity	"Ssz1 is not an ATPase in vitro, and even its ability to bind ATP is dispensable in vivo"	Yeast	Hig ort
PMID: 32198371	Functional/ structural	Qualifies BP as cotranslational	Substrate- engagement mode	Transient, low-affinity nascent-chain binding ensures flux through RAC/ Ssb	Yeast	Hig ort
PMID: 30635648	Functional (mechanistic)	Competing / qualifies nucleus	Nuclear/ chromatin role	Intracellular HSP70L1 maintains repressive H3K27me3/ H2AK119Ub1, inhibits Ash1l recruitment	Human dendritic cells	Me dis fro car

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Con & lim
 19946888	Localization (HDA)	Qualifies/ weak membrane	Plasma- membrane residence	Whole- membrane NK-cell proteome; HSP70s are classic co- fractionating contaminants	Human NK (YTS) cells	Low gen fra not
PMID: 14592822 , PMID: 21730052	Recombinant assay	Qualifies/ weak secretion	Extracellular/ membrane function	Recombinant HSP70L1 binds TLR4, matures DCs, Th1 adjuvant	Human DCs, recombinant protein	Low enc sec use pur pro
UniProt Q0VDF9 sequence	Computational (direct sequence)	Refutes canonical ATPase site	Catalytic-site integrity	PS00297 IDLGTTxS degenerate (GVHLGCTSA); no EEVD/ GGMP; no experimental PDB	In silico	Hig seq fac fun inf

GO Curation Implications (leads requiring curator verification)

GO term	Aspect	Current basis	Recommended action (lead)
GO:0016887 ATP hydrolysis activity	MF	IEA (IPR013126) + IBA (PTN000452648)	Remove / do not treat as core , or apply NOT-style scrutiny. Direct assay near background; hydrolysis-site mutants complement; ortholog structurally inert. Prefer GO:0005524 ATP binding as the supported MF.
GO:0042026 protein refolding	BP	IBA only	Generalize/replace with GO:0051083 de novo cotranslational protein folding (TAS, PMID16002468), the experimentally supported process. Refolding not directly supported.
GO:0005634 nucleus	CC	IBA only	Treat as non-core carry-over. A separate, primary chromatin function exists (PMID30635648) — curate that on its own evidence, not via the IBA.
GO:0005886 plasma membrane	CC	Stale IBA / absent from current path	Remove / do not annotate. No endogenous surface evidence; only generic membrane fraction + recombinant immunology.
GO:0005829 cytosol	CC	IDA ×2 (PMID16002468, PMID21231916)	Retain (core).
GO:0005840 ribosome	CC	IDA (PMID16002468)	Retain (core).
GO:0051083 de novo cotranslational protein folding	BP	TAS (PMID16002468)	Retain/promote as core BP.

The most informative MF for HSPA14 is **ATP binding (GO:0005524)** plus a co-chaperone / unfolded-protein-binding role in the ribosome-associated complex — not "ATP hydrolysis activity" and not merely "protein binding."

Mechanistic Scope

The immediate molecular function being tested is whether HSPA14's NBD **turns over ATP** and whether its SBD **refolds released denatured clients**. The evidence indicates neither: the NBD binds nucleotide but is catalytically inert (ortholog structures; degenerate human catalytic signature; near-background human assay), and the rudimentary SBD makes only transient, low-affinity contacts with **nascent** chains at the ribosome. The genuine downstream/pathway consequences frequently cited (TLR4/dendritic-cell adjuvant activity, breast-cancer prognosis, HIV replication modulation, prion suppression in yeast complementation) are context-specific phenotypes or recombinant-protein effects — they are not evidence of the intrinsic biochemical activities in the seed hypothesis. The chromatin/H3K27me3 role (PMID30635648) is a distinct, potentially direct intracellular function that must be evaluated separately, not folded into the phylogenetic nucleus IBA.

Conflicts and Alternatives

- **Paralog / family carry-over.** The strongest driver of the seed annotations is InterPro/PANTHER inheritance from the canonical, catalytically active Hsp70 branch. HSPA14 is an atypical Ssz1-type Hsp70; applying the family consensus over-annotates it. The degenerate PS00297 motif is the clearest paralog-discrimination signal.
- **Ortholog inference vs. direct human data.** All catalytic-inactivity *structures* are from fungal Ssz1; there is no experimental structure of human Q0VDF9. This is a species-inference limitation, but it is corroborated by the human sequence degeneracy and the near-background human assay, making the inference robust.
- **Recombinant vs. endogenous.** The extracellular/TLR4 immunology uses purified recombinant HSP70L1. Adjuvant activity of a purified protein does not establish endogenous secretion or plasma-membrane residence — a common source of CC over-annotation for HSP70-family proteins.
- **Co-fractionation artifact.** The membrane HDA is a whole-membrane proteome; HSP70s co-fractionate broadly. This is a classic false-positive route to membrane CC terms.

- **Negative-result logic.** The luciferase-refolding screen is negative for HSPA14; a curator should note this cannot exclude every substrate/cofactor context, but it does remove the direct basis for a refolding EXP annotation.

Limitations and Knowledge Gaps

1. **No experimental structure of human Q0VDF9.** Checked: UniProt lists only an AlphaFold model. Matters because human catalytic inactivity is inferred from ortholog structures + sequence. Resolution: a cryo-EM/crystal structure of human mRAC, or a rigorously controlled ATPase assay of highly purified human HSPA14 with proper background subtraction.
 2. **Assay sensitivity floor.** The ~0.01 ATP/min value sits near background; the study cannot formally exclude very slow hydrolysis. Resolution: single-turnover or high-sensitivity kinetics (e.g., malachite-green with rigorous controls, or NMR-based readouts).
 3. **Nuclear/chromatin function.** PMID30635648 reports a chromatin role, but whether HSPA14 itself localizes to chromatin (vs. acting indirectly) needs direct evidence. Resolution: endogenous ChIP/CUT&RUN and validated nuclear fractionation with tagged endogenous protein.
 4. **Plasma-membrane residence.** Only generic fraction + recombinant data exist. Resolution: surface biotinylation / non-permeabilized immunostaining of endogenous protein in relevant cells.
 5. **Substrate breadth of refolding.** The negative result is for luciferase/polyQ only. Resolution: broader client panel or unbiased interactome under denaturing/recovery conditions.
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Discriminating Tests (recommended)

- **High-sensitivity single-turnover ATPase assay** of purified human HSPA14 ± DNAJC2/MPP11, with the K68A/E172A hydrolysis-site mutants and the ATP-binding-deficient mutant as internal controls — directly tests intrinsic hydrolysis above background.
- **Cryo-EM of human mRAC on the ribosome** — resolves whether the human NBD can adopt the closed/catalytic conformation, testing the ortholog inference.

- **Endogenous surface biotinylation + non-permeabilized IF** in NK/dendritic cells — discriminates true plasma-membrane residence from co-fractionation.
 - **Endogenous nuclear fractionation + CUT&RUN** for tagged HSPA14 — tests direct chromatin engagement vs. indirect effect underlying PMID30635648.
 - **Broad refolding/holdase panel** (multiple denatured clients, aggregation-prevention vs. active refolding readouts) — tests whether any substrate class supports GO:0042026.
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Curation Leads (require curator verification)

1. **Downgrade/remove GO:0016887** (ATP hydrolysis activity) from core; retain **GO:0005524 ATP binding**. Lead references: PMID21245388 (snippet: *"ATP binding, but not ATP hydrolysis, by Hsp70L1 affected mRAC's function as a J-domain partner of Hsp70"*), PMID17901048 (*"Ssz1 is not an ATPase in vitro..."*), PMID28067917 (*"Ssz1 is catalytically inert..."*), PMID23202586.
 2. **Replace GO:0042026** (protein refolding) with **GO:0051083 de novo cotranslational protein folding** as core BP (TAS, PMID16002468); note the PMID21231916 negative luciferase result and family substrate-specificity.
 3. **Flag GO:0005634 nucleus** as non-core IBA carry-over; open a separate curation question on the PMID30635648 chromatin role (snippet: *"intracellular HSP70L1 inhibits the recruitment of Ash1l to and maintains the repressive H3K27me3 and H2AK119Ub1 modifications..."*).
 4. **Remove GO:0005886 plasma membrane**; document that it is absent from the current GO_Central path and supported only by generic membrane fraction (PMID19946888) and recombinant immunology (PMID14592822, PMID21730052).
 5. **Retain GO:0005829 cytosol and GO:0005840 ribosome** as core (IDA, PMID16002468/ PMID21231916).
 6. **Suggested curator questions:** Does any highly purified human HSPA14 preparation show hydrolysis above background? Is there any endogenous (non-recombinant) evidence for secretion/surface residence? Is the chromatin effect direct or indirect?
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Supported and Refuted Hypotheses Summary

Seed claim	Adjudication	Basis
Intrinsic ATP hydrolysis activity	Refuted / over-annotated	Near-background assay; hydrolysis-site mutants complement; degenerate motif; inert ortholog
Refolding of previously denatured proteins	Refuted as core / not directly supported	IBA-only; screen did not credit HSPA14; correct BP is cotranslational folding
Nuclear function	Unresolved (IBA carry-over refuted, but distinct chromatin lead exists)	PMID30635648 is a separate, potentially real function
Plasma-membrane function	Refuted / weakly supported	Absent from current path; generic fraction + recombinant only
ATP-binding mRAC cofactor / cotranslational folding	Supported (established core)	IDA/TAS; PMID16002468, PMID21245388

Prepared as a hypothesis-level curation lead. Structural conclusions about catalytic inactivity derive from the fungal ortholog Ssz1 and human sequence analysis; they are strong but inferential for human HSPA14 pending a direct human structure or high-sensitivity assay. All annotations flagged here require curator verification against current GO_Central provenance.